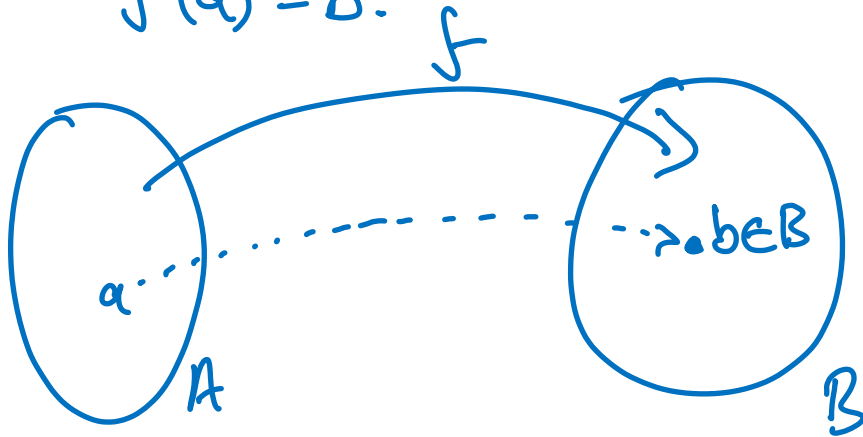


Lecture 5

Last lecture: Sets, functions, onto (surjective)

(surjective)

Defn: $f: A \rightarrow B$ is onto
if for every $b \in B$ there
exists an $a \in A$ such that
 $f(a) = b$.

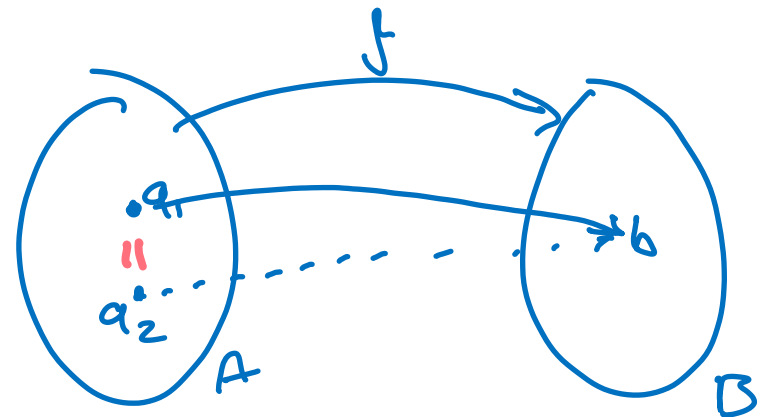


$$\forall b \in B (\exists a \in A \ f(a) = b).$$

" Every element of $b \in B$ is
the image of at least one
 $a \in A$. (existence of $b: f(a) = b$)

(injective)

Defn: $f: A \rightarrow B$ is one-to-one
if for every $a_1, a_2 \in A$
 $f(a_1) = f(a_2) \rightarrow a_1 = a_2$.



$$\forall a_1, a_2 \in A (f(a_1) = f(a_2) \rightarrow a_1 = a_2)$$

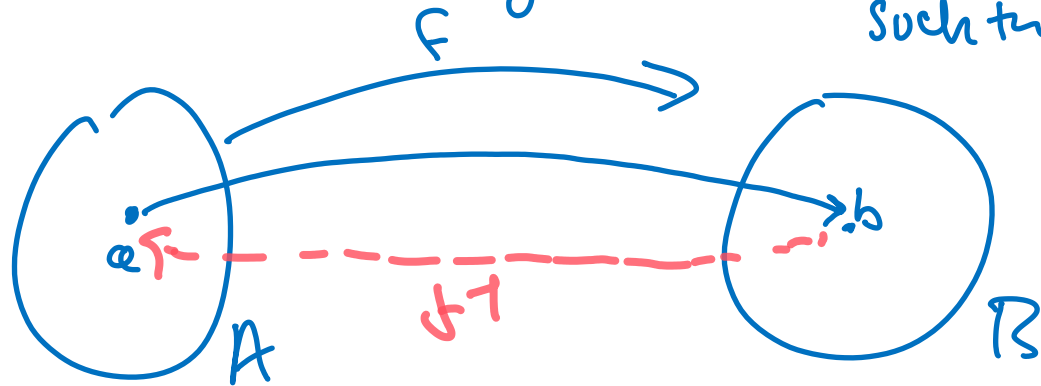
" Every element $b \in B$ is
the image of at most one
 $a \in A$. (uniqueness)

<u>ex:</u>	<u>A</u>	<u>B</u>	<u>rule</u>	onto	one-to-one
$f: \mathbb{Z} \rightarrow \mathbb{Z}$			$f(x) = x^2$	No. -1, 2	No, $1 = i^2 = (-i)^2$
$g: \mathbb{R} \rightarrow \mathbb{R}$			$g(x) = x^2$	No. -1.	No, $g(\pm x) = x^2$
$n: C \rightarrow C$			$n(x) = \begin{cases} \text{the person to the left of } x \\ \text{Nikhil if no such person.} \end{cases}$ (not a formula)	No.	No. b = Nikhil.
$m: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$			$m(a, b) = ab$	Yes.	No. 12.
$p: A \rightarrow \mathcal{P}(A)$			$p(a) = \{a\}$	No. $b = \emptyset, \{a_1, a_2\}$	Yes. $\{a_1\} = \{a_2\} \rightarrow a_1 = a_2$
$h: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$			$h(x) = x^2$	Yes.	No.
bijection $\Rightarrow$ $s: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ $s^{-1}: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$			$s(x) = x^2$ $s^{-1}(y) = +\sqrt{y}$	Yes	Yes.

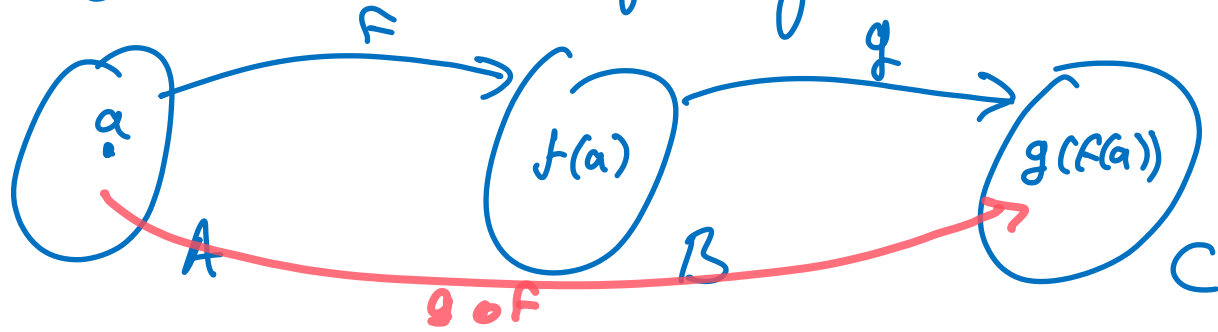
Defn: $f: A \rightarrow B$ is a bijection if it is both onto and 1-1.

Observe: Every $b \in B$ is the image of exactly one $a \in A$. In this case, we can define

Defn: $f^{-1}: B \rightarrow A$ by $f^{-1}(b) =$ the unique $a \in A$ such that $f(a) = b$



Defn: Given $f: A \rightarrow B$ and $g: B \rightarrow C$ their composition $g \circ f: A \rightarrow C$ is defined by $(g \circ f)(a) = g(f(a))$.



$g \circ f \neq f \circ g$

Prop 1: If $f: A \rightarrow B$ is onto and $g: B \rightarrow C$ is onto,
then $g \circ f: A \rightarrow C$ is onto.

Proof: Let f, g be arbitrary onto functions.

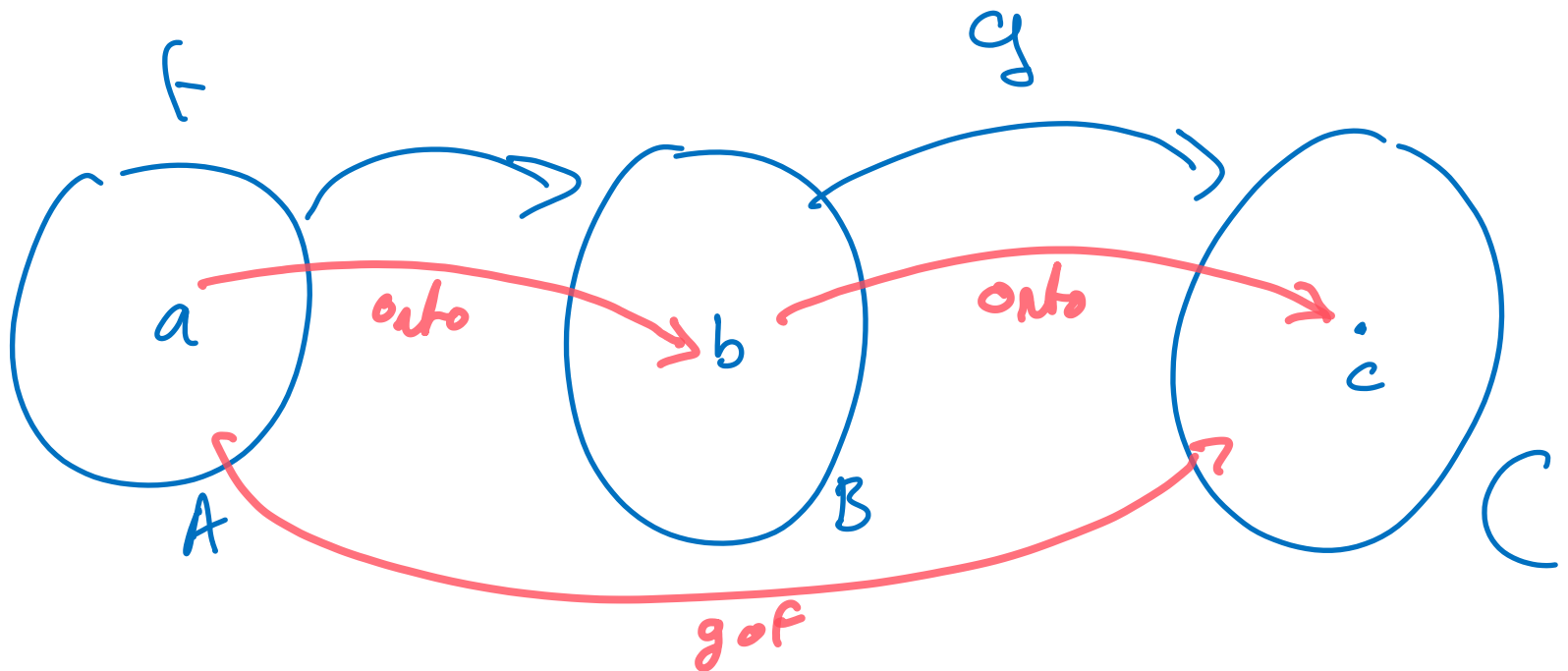
Let $c \in C$ be arbitrary.

Since g is onto, we can choose $b \in B$ s.t. $g(b) = c$.

Since f is onto, we can choose $a \in A$ s.t. $f(a) = b$.

Thus, $(g \circ f)(a) = g(f(a)) = g(b) = c$, as desired. $\square$

f, g onto.
 $c \in C$
 $b \in B$



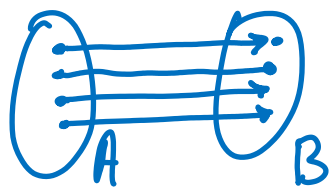
HW: f, g are 1-1 $\longrightarrow$ $g \circ f$ is 1-1
bijections $\longrightarrow$ $g \circ f$ is bijective.

II Cardinality

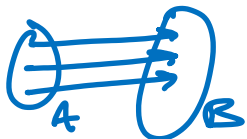
Defn: A set is finite if it has n elements for some $n \in \mathbb{Z}_{\geq 0}$. Its cardinality is n , denoted $|A| = n$.

Prop2 (Hw): If A, B are finite sets:

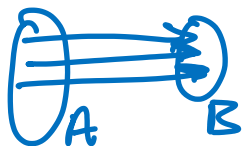
(cartoon)



$$|A| = |B| \iff \exists f: A \rightarrow B \text{ bijection.}$$



$$|A| \leq |B| \iff \exists g: A \rightarrow B \text{ one-to-one.}$$



$$|A| \geq |B| \iff \exists h: A \rightarrow B \quad \text{onto.}$$

Defn: Two sets A, B have the same cardinality ($|A| = |B|$) if $\exists f: A \rightarrow B$ bijection.

examples of infinite sets: $\mathbb{Z}_+, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$.

Defn: A function $f: \mathbb{Z}_+ \rightarrow A$ is called a sequence.

It can be written as $S = a_1, a_2, a_3, \dots, a_n, \dots$ where $a_n = f(n)$.

A sequence in which every element of A appears exactly once is called an enumeration of A .

ex: $1, 2, 3, 4, 5, \dots$ $f(n) = n$.

Defn: A set A is called countable if it is finite or $|A| = |\mathbb{Z}_+|$ i.e. $\exists f: \mathbb{Z}_+ \rightarrow A$ bijection
i.e. there is an enumeration of A .

An infinite set which is not countable is uncountable.

Prop 2: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\} \supseteq \mathbb{Z}_+$
is countable. (idea $|\mathbb{Z}| = |\mathbb{Z}_+|$)

Proof: Consider the sequence

$$S = 0, \overline{1}, \underline{-1}, \overline{2}, \underline{-2}, \dots$$

which alternates positive and negative integers.

Observe that every positive $n \in \mathbb{Z}$ appears exactly once (in position $2n$) and every negative $-n \in \mathbb{Z}$ appears exactly once (in position $2n+1$), and 0 appears exactly once. Thus S is an enumeration, as desired. $\square$

Alternative: Define $f: \mathbb{Z} \rightarrow \mathbb{Z}_+$ by $f(n) = \begin{cases} 2n & n > 0 \\ 2(-n)+1 & n < 0 \\ 0 & n = 0 \end{cases}$
(harder to read) Formally show 1-1 and onto.

Prop 3: $\mathbb{Z}_+ \times \mathbb{Z}_+ = \{ (a,b) : a,b \in \mathbb{Z}_+ \}$, is countable.
($|\mathbb{Z}_+| = |\mathbb{Z}_+ \times \mathbb{Z}_+|$)

Proof:

Consider the infinite array

(use pictures)

	$\nearrow D_2$	$\nearrow D_3$	$\nearrow D_4$	(or zigzag)
(1,1)	(1,2)	(1,3)	...	
(2,1)	(2,2)	(2,3)	...	
(3,1)	(3,2)	(3,3)	...	
...	...	...	...	

in which the j^{th} row is an enumeration of $\{j\} \times \mathbb{Z}_+$.

For $s \geq 2$, let $D_s = \{ (a,b) : a+b=s \}$ denote the diagonal of pairs with sum equal to s .

Since each D_s is finite, we can define

the sequence

(each pair appears ≤ 1 in D_{a+b}
because $a, b \geq 1$).

$t =$ listing of D_2 , listing of D_3 , listing of $D_4, \dots$

Observe that each $(a, b) \in \mathbb{Z}_+ \times \mathbb{Z}_+$ appears exactly
one in t , in the diagonal D_{a+b} .

Thus, t is an enumeration of $\mathbb{Z}_+ \times \mathbb{Z}_+$, as desired. $\square$

Remark: For any countable A, B $A \times B$ is countable.

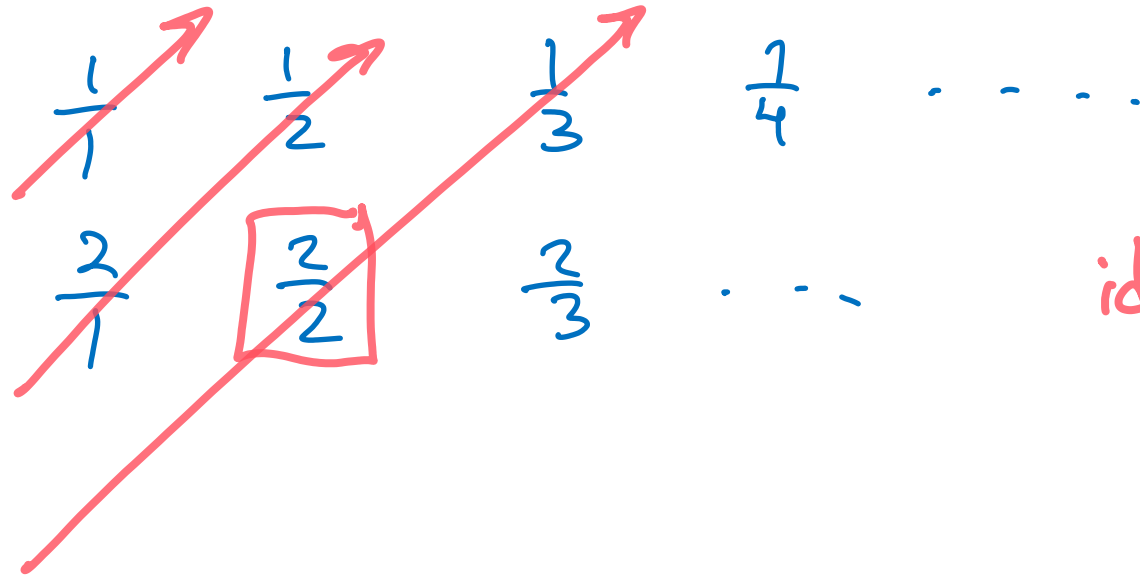
Proof Idea:

(Q) $A \times \underbrace{(A \times A)}_{\text{countable}}$

Remark: If A, B are countable, so is $A \cup B$:
 $a_1, b_1, a_2, b_2, a_3, b_3, \dots$
must delete repeated elements.

Prop 4: $\mathbb{Q}_+ := \left\{ \frac{p}{q} : p, q \in \mathbb{Z}_+, q \neq 0, \text{no common factors} \right\}$.
is countable. ($|\mathbb{Q}_+| = |\mathbb{Z}_+|$).

Proof: Consider the array



idea: delete
repeated
fractions.

Let s be the sequence obtained by listing the diagonals of this array:

$$s = s_1, s_2, s_3, \dots, s_n, \dots$$

Observe that every $p/q \in \mathbb{Q}_+$ appears at least once in s .

Define an element $s_n = p/q$ of s to be new if $s_n \neq s_m$ for all $m < n$.

Let $t = t_1, t_2, t_3, \dots$ where t_n is the n^{th} new element of s . By construction, every element of $\mathbb{Q}_+$ appears exactly once in t , as desired. $\square$

Remark: If $A \subseteq B$ and B is countable, A is countable. (delete repeated elements).

Defn: $(0,1) = \{0.d_1d_2d_3d_4\dots : d_i \in \{0,1,\dots,9\}, \text{ doesn't end in } 999\dots, \text{ not all zeros}\}$.

Thm (Cantor) : $(0,1)$ is uncountable.

($\forall f: \mathbb{Z}_+ \rightarrow (0,1)$, f is not a bijection).

Proof: Let $f: \mathbb{Z}_+ \rightarrow (0,1)$ be arbitrary.

We will show f is not onto. ($\exists x \in (0,1) : x \neq f(n) \forall n \in \mathbb{Z}_+$)

Consider the array

$$f(1) = 0.d_{11}d_{12}d_{13}\dots$$

$$f(2) = 0.d_{21}d_{22}d_{23}\dots$$

$$f(3) = 0.d_{31}d_{32}d_{33}\dots$$

$\vdots$

where d_{ij} is the j th digit of $f(i)$.

We will construct x such that $f(n) \neq x \forall n \in \mathbb{Z}_+$

$$\begin{aligned} f(1) &= 0.697343\dots \\ f(2) &= 0.569961234\dots \\ f(3) &= 0.694718882\dots \\ f(4) &= 0.555666321\dots \\ x &= 0.5555\dots \end{aligned}$$

Let $X = 0.X_1X_2X_3\dots$

$$\text{with } X_n = \begin{cases} 5 & \text{if } d_n \neq 5 \\ 6 & \text{if } d_n = 5 \end{cases}$$

different
from
 d_n

Let n be arbitrary. Observe $f(n) \neq X$
as they differ in the n th digit.

$\therefore f$ is not onto.

□